

Daily Logic Drill #1

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Statements & Quantifiers ($\forall/\exists$) · Negating Compound Statements · Negating Quantified Statements · Order of Quantifiers.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Apply a negation rule immediately — fluency, not first contact.

1. Negate: “For all integers n , $n^2 \geq 0$.”
2. Negate: “There exists a real number x such that $x^2 = -1$.”
3. Negate: “ n is even and n is prime.”
4. Negate: “ n is a multiple of 4 or n is a multiple of 6.”
5. True or false: “The negation of ‘for all x , $P(x)$ ’ is ‘for all x , not $P(x)$ ’.”
6. True or false: “not(P and Q) is the same as (not P) or (not Q).”
7. Negate: “ $x > 3$ and $x < 10$.”
8. Is the statement “some prime numbers are even” a $\forall$ -statement or a $\exists$ -statement in form?
9. True or false: “ $\exists x \forall y : x + y = 0$ ” and “ $\forall y \exists x : x + y = 0$ ” (both over the reals) mean the same thing.
10. Negate: “ n is prime or $n = 1$.”

Section B Manipulation Drills

(≤50 s each)

Negate nested and multi-part statements; start noticing where quantifier order changes meaning.

1. Consider the statement about Fred: “Every day next week, Fred will do at least one maths problem.” State the negation. (*after TMUA Specimen Paper 2 Q9*)
2. True or false: “not(P or Q or R) is the same as (not P) and (not Q) and (not R).”
3. Negate: “ n is a multiple of 2 and n is a multiple of 3 and n is a multiple of 5.”

4. Negate: “For every real $x > 0$, there exists a real $y > 0$ with $y < x$.”
5. True or false: “‘ $\forall n \exists m : m > n$ ’ and ‘ $\exists m \forall n : m > n$ ’ (both over the integers) mean the same thing.”
6. Negate the nested statement: “There exists a prime p such that for every prime $q > p$, q is odd.” Is the original statement true or false?
7. Negate: “ n is a multiple of 3, or (n is a multiple of 2 and n is a multiple of 5).”
8. True or false: “not(A and (B or C)) is the same as (not A) or ((not B) and (not C)).”
9. Negate: “ x is rational, or (x is irrational and $x > 0$).”
10. Negate: “For every prime $p > 2$, p is odd.” Is the negation true or false?

Section C Substitution & Structure

(≤ 90 s each)

Multi-step negations, nested quantifiers, and your first ordering-of-quantifiers MCQ.

1. For real x and positive integer n , consider: I. For all x and for all n , $x^2 < n$. II. For all x , there exists n such that $x^2 < n$. III. There exists x such that for all n , $x^2 < n$. Which of I, II, III are true? (after TMUA 2022 Paper 2 Q10)
2. Determine the complete set of real values of k for which the following is true: “For all real x , if $x < k$, then $x^2 > k$.”
 - A) no real numbers
 - B) $k < 0$
 - C) $k \leq 0$
 - D) $k \leq 1$
 - E) all real numbers
 - F) $k > 0$
3. Which is the correct negation of “For all primes p , p is odd or $p = 2$ ”?
 - A) There exists a prime p such that p is odd or $p = 2$.
 - B) There exists a prime p such that p is not odd and $p \neq 2$.
 - C) For all primes p , p is not odd and $p \neq 2$.
 - D) There exists a prime p such that p is not odd or $p \neq 2$.
4. Negate: “There exists a positive integer n such that $n^2 + n + 1$ is even.” Is the negation true or false?
5. Negate: “There exists a positive integer n such that for every positive integer $k \leq 10$, k divides n .” Is the original statement true or false?

6. Consider the statement about a real number x : “There exists a real number y such that $xy - x + y$ is positive.” For how many real values of x is this true? (after TMUA 2022 Paper 2 Q13)
- A) no values of x
 - B) exactly one value of x
 - C) all except exactly one value of x
 - D) all except exactly two values of x
 - E) all values of x
7. Negate: “There exists a real number x such that for every real number y , $xy = y$.” Is the original statement true or false?
8. Negate: “For every positive integer n , there exists a positive integer $m < n$ that is prime.” Consider the cases $n = 1$ and $n = 2$ carefully. Is the original statement true or false?

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Consider the statement: “For every finite sequence of positive integers, there exists a positive integer m such that every term of the sequence is at most m .” Which of the following is the correct negation?
- A) There exists a finite sequence of positive integers such that, for every positive integer m , some term exceeds m .
 - B) There exists a finite sequence such that, for every positive integer m , every term is at most m .
 - C) For every finite sequence, there exists a positive integer m such that some term exceeds m .
 - D) There exists an infinite sequence such that no positive integer bounds all terms.
2. Determine, with brief justification, whether the following is true or false: “For every positive real number ε , there exists a positive integer N such that for every integer $n \geq N$, $\frac{1}{n} < \varepsilon$.”
3. A selection S of n terms is taken from the arithmetic sequence $2, 5, 8, 11, \dots, 92$. What is the smallest value of n for which S must contain two distinct terms summing to 94? (after TMUA 2022 Paper 2 Q8)
- A) 15
 - B) 16
 - C) 17
 - D) 18
 - E) 24
 - F) 31
4. Statement (*): “For every finite set of primes $\{p_1, \dots, p_k\}$, there exists a prime not in the set.”
- (a) Negate (*). (b) Is (*) true or false? Justify briefly.

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$. Consider: I. “For every x , there exists y such that $f(y) = x$.” II. “There exists y such that for every x , $f(y) = x$.” (a) Are I and II logically equivalent? (b) Give a concrete function f for which I is true but II is false.

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths